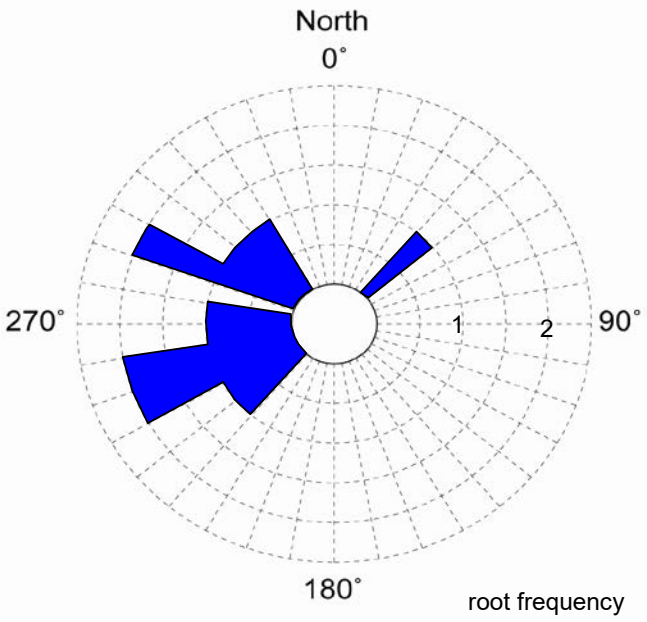
Sheet1 Ripples

	A	B	C	D	E	F	G	H
1	Data (θ)	sin θ	cos θ		Bin	Frequency	Frequency$^{1/2}$	
2	46	0.72	0.69		0			
3	241	-0.87	-0.48		10			
4	290	-0.94	0.34		20			
5	260	-0.98	-0.17		30			
6	251	-0.95	-0.33		40	1	1	
7	249	-0.93	-0.36		50			
8	278	-0.99	0.14		60			
9	295	-0.91	0.42		70			
10	297	-0.89	0.45		80			
11	316	-0.69	0.72		90			
12	314	-0.72	0.69		100			
13	326	-0.56	0.83		110			
14	309	-0.78	0.63		120			
15	254	-0.96	-0.28		130			
16	251	-0.95	-0.33		140			
17	224	-0.69	-0.72		150			
18	259	-0.98	-0.19		160			
19	238	-0.85	-0.53		170			
20	248	-0.93	-0.37		180			
21	sum	-14.86	1.17		190			
22					200			
23	**θ**	**-86**			210			
24	Mean (0-360 degrees)	**274**			220	1	1.0	
25					230	1	1.0	
26					240	3	1.7	
27	**circular variance**	**0.22**			250	4	2.0	
28					260	1	1.0	
29					270	1	1.0	
30					280			
31					290	3	1.7	
32					300	1	1.0	
33					310	2	1.4	
34					320	1	1.0	
35					330			
36					340			
37					350			
38					360			
39								
40								
41								
42								
43								
44								

	I	J	K	L	M
1					
2					
3	FORMULAS				
4	*xi* =SIN(θ*PI()/180)				
5	*yi* =COS(θ*PI()/180)				
6					
7					
8	θ=ATAN(*xr*/*yr*)*180/PI()				
9					
10	if x_r is positive and y_r is positive	add 0^0			
11	if x_r is positive and y_r is negative	add 180^0			
12	if x_r is negative and y_r is negative	add 180^0			
13	if x_r is negative and y_r is positive	add 360^0			
14					
15	circular variance=1-((SQRT(*xr*^2+*yr*^2))/n)				
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40	YOU NEED TO PLOT THE ROSE DIAGRAM BY HAND ON THE PRACTICAL				
41	HANDOUT- YOU DO NOT MAKE THE ROSE DIAGRAM IN EXCEL				
42					
43					
44					



	A	B	C	D	E	F	G
1	Data (θ)	sin θ	cos θ		Bin	Frequency	Frequency$^{1/2}$
2	242	-0.88	-0.47		0		
3	239	-0.86	-0.52		10		
4	240	-0.87	-0.50		20		
5	253	-0.96	-0.29		30		
6	235	-0.82	-0.57		40		
7	258	-0.98	-0.21		50		
8	272	-1.00	0.03		60		
9	239	-0.86	-0.52		70		
10	88	1.00	0.03		80		
11	175	0.09	-1.00		90		
12	200	-0.34	-0.94		100		
13	31	0.52	0.86		110		
14	sum	-5.96	-4.08		120		
15					130		
16	**θ**	**55.58**			140		
17	Mean (0-360 degrees)	**236**			150		
18					160		
19					170		
20	**circular variance**	**0.40**			180		
21					190		
22					200		
23					210		
24					220		
25					230		
26					240		
27					250		
28					260		
29					270		
30					280		
31					290		
32					300		
33					310		
34					320		
35					330		
36					340		
37					350		
38					360		
39							
40							
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Sheet 3 Aligned Shells

	A	B	C	D	E	F	G	H	I	J	K
1	Data (θ)	sin θ	cos θ		Data (θ)	sin θ	cos θ		Bin	Frequency	Frequency$^{1/2}$
2	75	0.97	0.26		75	0.97	0.26		0		
3	255	-0.97	-0.26		75	0.97	0.26		30		
4	180	0.00	-1.00		0	0.00	1.00		60		
5	25	0.42	0.91		25	0.42	0.91		90		
6	170	0.17	-0.98		170	0.17	-0.98		120		
7	195	-0.26	-0.97		15	0.26	0.97		150		
8	50	0.77	0.64		50	0.77	0.64		180		
9	250	-0.94	-0.34		70	0.94	0.34		210		
10	130	0.77	-0.64		130	0.77	-0.64		240		
11	20	0.34	0.94		20	0.34	0.94		270		
12	25	0.42	0.91		25	0.42	0.91		300		
13	5	0.09	1.00		5	0.09	1.00		330		
14	175	0.09	-1.00		175	0.09	-1.00		360		
15	190	-0.17	-0.98		10	0.17	0.98				
16	335	-0.42	0.91		155	0.42	-0.91				
17	180	0.00	-1.00		0	0.00	1.00				
18	195	-0.26	-0.97		15	0.26	0.97				
19	195	-0.26	-0.97		15	0.26	0.97				
20	10	0.17	0.98		10	0.17	0.98				
21	70	0.94	0.34		70	0.94	0.34				
22	150	0.50	-0.87		150	0.50	-0.87				
23	185	-0.09	-1.00		5	0.09	1.00				
24	180	0.00	-1.00		0	0.00	1.00				
25	5	0.09	1.00		5	0.09	1.00				
26	25	0.42	0.91		25	0.42	0.91				
27	205	-0.42	-0.91		25	0.42	0.91				
28	210	-0.50	-0.87		30	0.50	0.87				
29	25	0.42	0.91		25	0.42	0.91				
30	40	0.64	0.77		40	0.64	0.77				
31	190	-0.17	-0.98		10	0.17	0.98				
32	190	-0.17	-0.98		10	0.17	0.98				
33	180	0.00	-1.00		0	0.00	1.00				
34	35	0.57	0.82		35	0.57	0.82				
35	20	0.34	0.94		20	0.34	0.94				
36	5	0.09	1.00		5	0.09	1.00				
37	260	-0.98	-0.17		80	0.98	0.17				
38	185	-0.09	-1.00		5	0.09	1.00				
39	200	-0.34	-0.94		20	0.34	0.94				
40	175	0.09	-1.00		175	0.09	-1.00				
41	230	-0.77	-0.64		50	0.77	0.64				
42	235	-0.82	-0.57		55	0.82	0.57				
43	25	0.42	0.91		25	0.42	0.91				
44	45	0.71	0.71		45	0.71	0.71				
45	245	-0.91	-0.42		65	0.91	0.42				
46	50	0.77	0.64		50	0.77	0.64				
47	sum	1.67	-5.99		sum	18.75	26.14				
48											
49	**θ**	**-15.56**			**θ**	**35.65**					
50	Mean (0-360 degrees)	**164**			Mean (0-360 degrees)	**36**					
51											
52											
53	**circular variance**	**0.86**			**circular variance**	**0.29**					
54											